

Optimal Athletic Programming in Triathlon using Linear and Integer Programming

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September 20, 2026

Three decisions, one athlete

A competitive triathlete must decide:

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Model 1 (LP) $\rightarrow$ **Model 2 (0–1 MIP)** $\rightarrow$ **Model 3 (LP)**
training load race calendar pacing strategy

Coupled: Model 1's fitness state gates race readiness and sets the race-day energy budget.

Real data: 15 months of my own training

- Strava export: **598 activities**, 574 kept (Oct 2023 – Jun 2026)
- Per activity: Banister training impulse
 $\text{TRIMP} = t \cdot \text{HRr} \cdot 0.64 e^{1.92 \text{HRr}}$
- Exponential smoothing $\Rightarrow$ fitness CTL ($\tau=42\text{d}$),
fatigue ATL ($\tau=7\text{d}$)
- Current state: CTL = 78.1, ATL = 91.5,
TSB = -13.4 (fit but fatigued)
- Intensity buckets vs. *sport-specific* threshold HR
(run 173 / bike 166 / swim 164 bpm)

r_{db}/h	easy	mod	hard
swim	43.5	94.1*	136.6*
bike	68.3	86.9	142.7*
run	81.6	102.8	166.0*

* theoretical (sparse data)

Model 1 — training-load allocation (LP)

Variables: hours $h_{dbw} \geq 0$ per discipline $\times$ bucket $\times$ week ($3 \times 3 \times 16 = 144$)

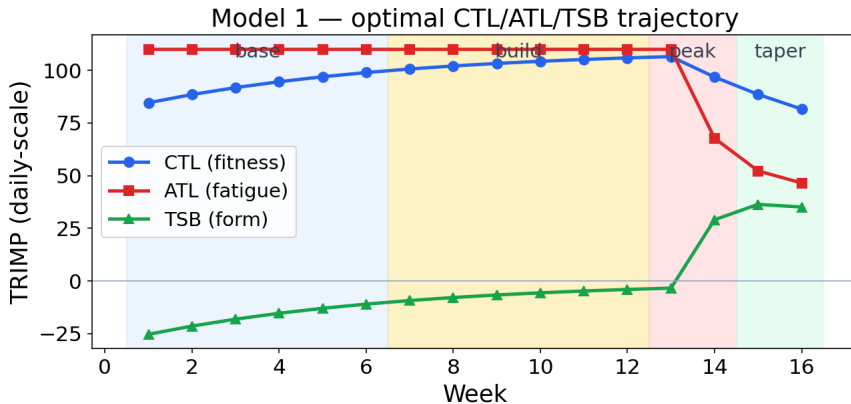
Banister dynamics as linear equalities (exact — smoothing is linear in load):

$$L_w = \sum_{d,b} r_{db} h_{dbw}, \quad \text{CTL}_w = \lambda_c \text{CTL}_{w-1} + (1-\lambda_c) \frac{L_w}{7}, \quad \text{ATL}_w = \lambda_a \text{ATL}_{w-1} + (1-\lambda_a) \frac{L_w}{7}$$

Objective: race-day performance $\max p_T = k_1 \text{CTL}_T - k_2 \text{ATL}_T$ (TSB-max alone would be degenerate: optimum = rest!)

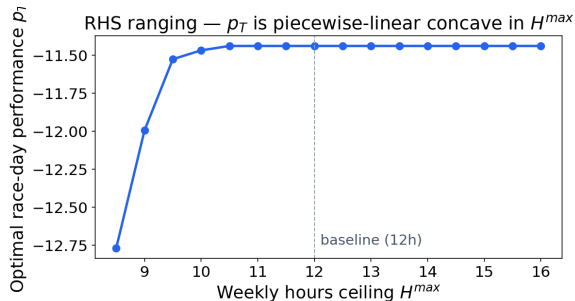
Constraints: weekly hour ceilings (phase-dependent) $\cdot$ discipline min/max $\cdot$ 80/20 polarization $\cdot$
 $\text{ATL} \leq 110 \cdot \text{ramp} \leq +10\%/\text{week} \cdot \text{taper caps}$

Model 1 — the optimizer discovers periodization



Rides the fatigue ceiling for 13 weeks (CTL 78 → 107), then tapers. Only weeks 15–16 are *constrained* to taper — **week 14 tapers by choice**. Race day: TSB = +35.

Model 1 — duality in action



RHS ranging (hours ceiling 8–16h):

- value function piecewise-linear, concave; slopes $1.55 \rightarrow 0.94 \rightarrow 0.12 \rightarrow 0$
- saturates at **10.5 h/week**: the ATL ceiling binds first

$\Rightarrow$ *recovery capacity, not free time, limits this athlete*

Largest shadow prices: race-week minimums (-12.9 /forced run hour in week 16) — the model wants full rest

Model 2 — race-calendar selection (0–1 MIP)

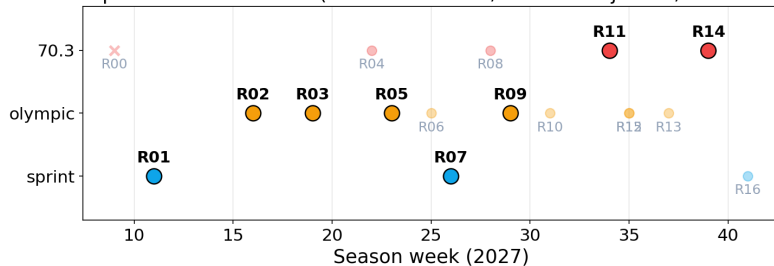
17 candidate races (Greek / European / long-haul), binary x_i

$$\max \sum_i q_i \underbrace{(\alpha \text{ pts}_i + \beta \text{ strat}_i + \gamma \text{ enj}_i + \varphi \text{ fit}_i)}_{\text{weather-risk-adjusted value}} x_i$$

- **Two-dimensional knapsack:** $\sum \kappa_i x_i \leq 2500$ EUR and $\sum \delta_i x_i \leq 12$ vacation days
- recovery spacing $x_i + x_j \leq 1$ (1/2/4 weeks by distance class); ≤ 2 races/month; ≥ 1 A-race
- **readiness gate from Model 1:** projected CTL must reach the class requirement — excludes a high-value early-season 70.3
- suitability fit_i from *objective* course data: elevation gain/km vs. athlete's flat-course preference

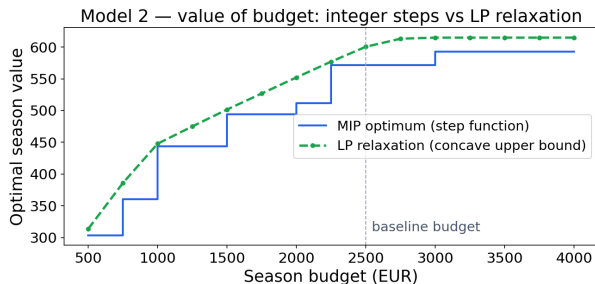
Model 2 — optimal season

Model 2 — optimal race calendar (solid = selected, faded = rejected, x = not race-ready)



8 of 17 races, value 571.7, **2250/2500 EUR and 11/12 days used**. Recovery forces National Champs vs. home 70.3 — optimizer picks the 70.3.

Model 2 — why Branch & Bound is needed



- LP relaxation: $z = 600.5$ vs. MIP 571.7 — **integrality gap 5.0%**
- 8 fractional variables; the champs/70.3 conflict splits exactly **0.5/0.5**
- MIP value-of-budget is a **step function** under the concave LP curve
- above 3000 EUR the *vacation-days* budget binds instead — the scarce resource switches
- **clique cuts** (Bron–Kerbosch on the conflict graph): 4 valid inequalities $\sum_{i \in C} x_i \leq 1$ tighten the relaxation $600.5 \rightarrow 591.0$, closing a third of the gap (**5.04% $\rightarrow$ 3.38%**) without touching the integer optimum

Model 2 — the dual problem

LP relaxation is already in standard form $\max\{c^T x \mid Ax \leq b, x \geq 0\}$, so the dual is $\min\{b^T y \mid A^T y \geq c, y \geq 0\}$:

$$\kappa_i y_B + \delta_i y_D + y_{\text{month}(i)} + \sum_j y_{ij} - [i \in A] y_A \geq v_i$$

a race enters only if its value covers the shadow cost of what it consumes

Dual	Meaning	Price
$y_{13,14}$	Hellenic Champs vs. Costa Navarino conflict	56.40
$y_{06,07}$	Hamburg vs. Thessaloniki conflict	31.67
y_B	one extra euro	0.094
y_D	one extra vacation day	≈ 0

⇒ the scarce resource is neither money nor days off but **calendar space**: recovery between races.

Model 2 — network-flow extension

Drop the knapsacks and monthly cap $\Rightarrow$ only recovery spacing remains:

feasible calendars = paths in a DAG (nodes: races in week order; arc $i \rightarrow j$ iff $\text{gap} \geq \text{recovery}(i)$)

Three methods, identical result ($z = 639.4$):

- 1 dynamic programming (longest path, topological order)
- 2 **flow LP with continuous variables** — solution integral anyway: network matrix $\Rightarrow$ *totally unimodular*
- 3 restricted 0–1 MIP

Node–arc incidence matrix $\Rightarrow$ every basis is a spanning tree: exactly the class *Network Simplex* is designed for (18 nodes, 136 arcs). $\Rightarrow$ the knapsack constraints are precisely what *breaks* the network structure and forces Branch & Bound in the full model.

Model 3 — race pacing (LP with linearization)

Variables: minutes $t_{\ell z}$ per leg $\times$ intensity zone (Z1–Z5); 15 variables

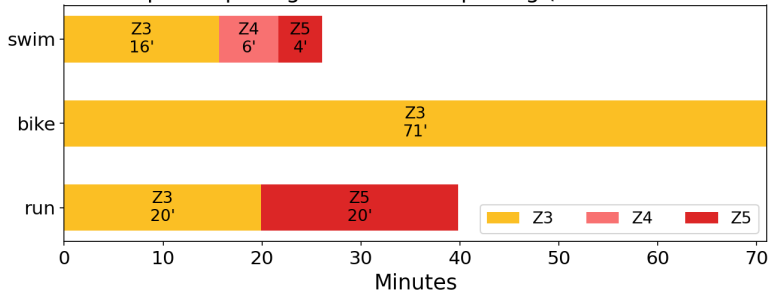
Linearization: within a zone, speed and energy rate are constants — bike speed $\propto \phi(\gamma) \cdot (\text{power})^{1/3}$, economy degrades with intensity. $\phi(\gamma)$ is the course-gradient factor from the cycling power balance (harmonic mean of up/down speeds at constant power) — **derived, not fitted**

$$\min \sum_{\ell, z} t_{\ell z} + T_1 + T_2 \quad \text{s.t.} \quad \underbrace{\sum_z s_{\ell z} t_{\ell z} \geq d_\ell}_{\text{distances}}, \underbrace{\sum e_{\ell z} t_{\ell z} \leq k_E \cdot \text{CTL}_T}_{\text{energy budget (Model 1)}}, \underbrace{\text{Z4+ time caps}}_{\text{per leg}}, \underbrace{\varphi \cdot G}_{\text{bike} \rightarrow \text{run}}$$

The coupling: each kJ of hard bike work reduces effective run coverage.

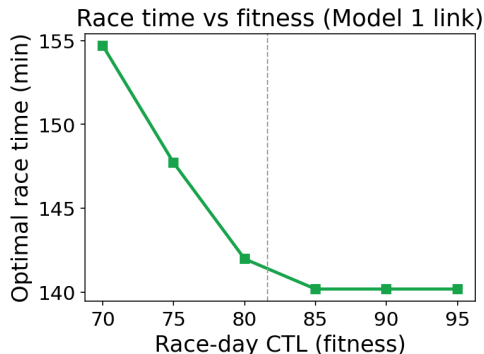
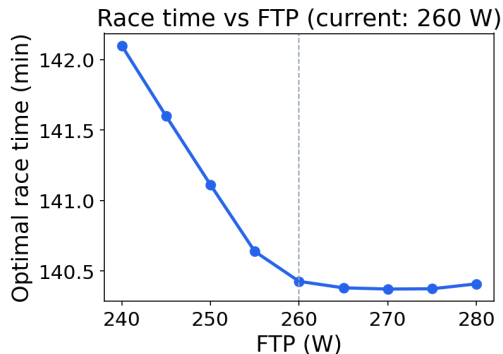
Model 3 — “steady bike, hard run,” derived

Model 3 — optimal pacing: time in zone per leg (total 137' + transitions)



Olympic distance in 140.4 min; 9389 kJ budget **exactly** exhausted. Bike entirely at Z3: hard biking buys little speed (cube root) and drains the run.

Model 3 — what is a watt worth?



Energy dual: $1 \text{ kJ} = 0.43 \text{ s}$. Below 260 W each watt $\approx 6 \text{ s}$; **at the current FTP the value collapses** — with a fixed energy budget, more power cannot be fueled. One CTL point = 69–84 s up to CTL ≈ 85 . $\Rightarrow$ *build endurance, not peak power.*

Model 3 — validated against three real races

Race	CTL	Swim		Bike		Run	
		mdl	act	mdl	act	mdl	act
Spetsathlon '26 (sprint*)	64.9	12.9	13.2	47.3	50.9	18.7	17.7
Epidavros '25 (Olympic)	55.6	27.2	25.8	88.8	91.5	38.9	42.0
70.3 Costa Navarino '25 (fueled)	71.4	33.3	32.9	184.6	180.3	86.4	115.2

Race days found in the raw export (the only three with swim+bike+run); fitness from the pipeline. * non-standard course.

- swim within 0.5 min every time; measured thresholds work
- **binding constraints switch with distance**: speed caps (sprint) → energy (Olympic) → *infeasible unfed* (70.3)
- calibrated fuelling $r \approx 21$ kJ/min = **75 g carbs/h** — **the standard guideline**, recovered from the race
- the 70.3 run “wall” (4:53 → 5:52/km, HR 162 → 145): the energy constraint violated in real life

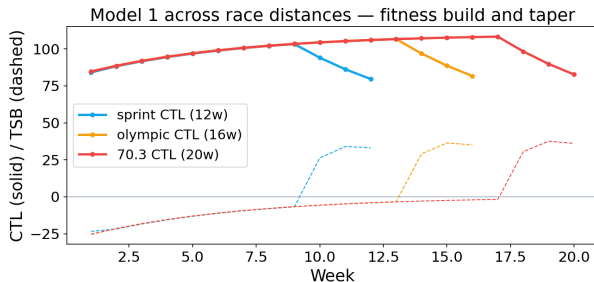
What the bike residual is made of

The bike is the least accurate leg. Is the residual noise, or a missing term?

Bike leg	m/km	no gradient	$\phi(\gamma)$	empirical fit	actual
Spetsathlon (24.4 km)	15.2	43.3	47.3	55.0	50.9
Epidavros (38.7 km)	15.5	78.9	88.8	87.7	91.5
Costa Navarino (88.5 km)	9.1	177.5	184.6	180.0	180.3
total error		23.0 min	10.6 min	8.2 min	
fitted to these races		0	0	2	

- ignore gradient $\Rightarrow$ 23 min error and **too fast on every course** — a missing term, not noise
- $\phi(\gamma)$ from physics: 10.6 min, errors on **both** sides
- a 2-parameter fit *on* these races buys only 2.4 min more — so gradient explains essentially all of the recoverable error
- residual sign is consistent: ϕ assumes threshold power on the descents, which nobody does $\Rightarrow$ still slightly fast on the two hilly courses

Sensitivity across problem sizes and parameters



Distance scenarios (12/16/20 weeks, 144–240 vars): longer horizons help (p_T : $-13.5 \rightarrow -10.4$); build–taper structure invariant

k_2 **fatigue weight** $\in [1, 3]$: race-day TSB stays 35 ± 2 — taper depth is *constraint*-determined (race-week minimums), robust to the objective's calibration

Cost ranging (c -perturbation): plan invariant for $k_1 \in [0.68, 1.56]$, $k_2 \in [1.29, 2.95]$ — robust to the least measurable parameters

Sizes/times: LP 192 vars / MIP 17 bins / LP 15 vars; all <10 ms (CBC/Simplex)

Integrating the pipeline: coherence + fixed point

Do the objectives agree? Model 1 maximizes $p_T = k_1\text{CTL}_T - k_2\text{ATL}_T$; the athlete wants *race time*.

- default budget $E = k_E\text{CTL}$ ignores ATL $\Rightarrow$ four different plans all give **the same 140.2 min** — Model 3 is blind to training
- form-based $E = E_{\text{base}} + k_E(\text{CTL}_T - \lambda\text{ATL}_T)$ is affine in p_T **iff** $\lambda = k_2/k_1 \Rightarrow \arg \max p_T = \arg \min \text{time}$
- $E > 0$ needs $\lambda < \text{CTL}_T/\text{ATL}_T = 1.75$, so the assumed $k_2 = 2$ is *incoherent*; cost ranging says the plan is invariant on $[1.29, 2.95]$, so $\lambda = 1.5$ fixes it **changing no variable**

Which comes first? The A-race sets Model 1's macrocycle; Model 1's load drives Model 2's readiness gate. Iterate instead of assuming:

Iter.	Macrocycle	Weeks	Selected A-race	Outcome
1	olympic	16	R14 Costa Navarino (70.3)	re-plan as 70.3
2	70.3	20	R14 Costa Navarino (70.3)	fixed point

Converges in 2 steps, and exposes that the calendar's 70.3 A-race needs a **20-week** build — not the 16-week Olympic one originally assumed.

What the models actually recommend

- **Train more, much easier.** Actual: 6.6 h/wk at **37%** hard. Optimal: 9.0 h/wk at **15%**. Anti-polarised — the one change with the largest payoff.
- **Buying training time is worthless** above 10.5 h/wk (zero shadow price); *recovery capacity* is what binds.
- **Build endurance, not watts.** +1 CTL = 69–84 s; +1 W $\approx$ 6 s below 260 W, $\approx$ 0 above.
- **Ride Z3, run hard, pace evenly.** The 70.3 positive split (4:53 $\rightarrow$ 5:52/km, HR 162 $\rightarrow$ 145) cost $\approx$ 29 min.
- **Fuel 75 g/h**; gut training worth $\approx$ 1.7 min per kJ/min — at 70.3 only.
- **Protect the calendar, not the budget.** Recovery conflict dual **56.4** vs. 0.094 per euro.
- **Skip early-season Dubai** (not fit in time); if the A-race is a 70.3, build **20 weeks**, not 16.

- Three coupled models on **real personal data**, spanning the syllabus: LP + Simplex, duality & sensitivity, IP + Branch & Bound, network flows + total unimodularity
- The optimizer **re-derives coaching wisdom**: periodization, polarized intensity, “steady bike, hard run”
- Duals price the athlete’s true bottlenecks: *recovery capacity* (training), *money* (calendar), *endurance* — *not watts* (race day)
- Fully reproducible: `python -m src.run_all`, 59 tests
<https://github.com/manos1503/triathlon-optimization>

Thank you! Questions?

Backup: limitations — stated honestly

- **Weekly time step** for Banister dynamics (daily would shift constants, not structure)
- **Sequential pipeline**, not joint optimization — co-optimizing training + calendar is a natural (larger) MIP extension
- **Constructed parameters** in Models 2–3 (race values, zone speeds, k_E) — documented assumptions; sensitivity analysis shows which conclusions survive perturbation
- LP's +35 TSB over-taper traced to race-week minimum constraints, not to the model concept

Backup: alternative optima & heuristic benchmark

Alternative optima (degeneracy made visible): fix $p_T = p^*$, re-solve min/max total hours

same performance with **124.2h** or **162.3h** of season training — a 38h efficiency gap

vs. standard coaching heuristic (fill available hours, 20/45/35 split, 80/15/5 intensity):

- violates the fatigue ceiling in **13 of 16 weeks** (chronic overtraining)
- $p_T = -69.5$ vs. optimal -11.4
- but: heuristic race-day TSB $+21$ is inside the coaching band $+15..25$; the LP's $+35$ is not → see k_2 sensitivity